[01SQHOV] - TECHNOLOGIES FOR AUTONOMOUS VEHICLES

Assignment 1: Global path planning

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***Abstract*—This report presents a comparative analysis of Dijkstra's algorithm and A* with three heuristic functions: Euclidean, Manhattan and Haversine on real road networks. The graph was obtained from OpenStreetMap API via OSMnx library, using the cities of Turin and Aosta. The two cities differ significantly in scale and structure. Edge weights were defined as travel time required to travel across the road (edge) expressed in seconds. A key challenge was ensuring unit consistency between the heuristic estimates, which operate on geographic coordinates, and the edge weights expressed as time. This report analyzes different scaling factors and their effect on admissibility and search efficiency. Results show that the Haversine heuristic, which computes the actual great-circle distance, provides the tightest admissible bound and reduces explored nodes by 30–46% compared to Dijkstra. The Euclidean and Manhattan distances, operating on raw latitude/longitude coordinates, require scaling to account the non-uniform relationship between degrees and meters, and Manhattan in particular is difficult to make simultaneously tight and admissible on geographic coordinates.**

I. INTRODUCTION

Global path planning is a core component of autonomous vehicle navigation. Given a road network graph, the task is to find the shortest (or fastest) path between two nodes. Dijkstra's algorithm solves this optimally by expanding nodes in order of increasing cost. It is optimal and complete, but it explores nodes uniformly in all directions, which can be wasteful when the destination is known in advance. A* improves on this by adding a heuristic component which estimates the remaining cost, biasing the search toward the destination.

The effectiveness of A* relies entirely on the heuristic: a weak one makes it degenerate in Dijkstra, while an overly optimistic one may speed up the search at cost of optimality. On geographic road networks, the main difficulty is that heuristic distance functions operate on latitude/longitude coordinates, while edge weights are in metric units.

In order to complete the assignment, it was necessary to implement both algorithms on road networks of Turin (11,668 nodes, 25,060 edges) and Aosta (817 nodes, 1,712 edges), extracted from OpenStreetMap via OSMnx. A* evaluation was made evaluating three different heuristic functions: Euclidean, Manhattan and Haversine. The central challenge, as it will be discussed later, was that geographic coordinates in degrees do not directly correspond to the metric units used in graph edge weights, and reconciling these two systems is non-trivial. To assess the challenge, a scaling factor was introduced to analyze the trade-off between admissibility and pruning power.

II. METHODS

A. Graph construction

Road networks were downloaded from OpenStreetMap via OSMnx library. The downloaded graph was then filtered by retaining only the largest strongly connected components. This step removes discnnected road segments (dead-end service roads, isolated parking areas and related) This ensures full reachability between any pair of nodes.

Edge weights represent travel time in seconds:

$$w(e) = \frac{\text{length}(e)}{\text{maxspeed}(e)/3.6} \quad (1)$$

where length is in meters and maxspeed in km/h (divided by 3.6 to convert to m/s). Missing speed limits default to 40 km/h.

B. Algorithms

Both Dijkstra and A* use a min-heap priority queue with lazy deletion. A* uses priority $f(n) = g(n) + h(n)$, where $g(n)$ is the cost from source to end and $h(n)$ the heuristic estimate of the remaining cost from n to destination. For optimality, $h(n)$ must be admissible: it must never overestimate the true remaining cost.

C. Heuristics and scaling

All three heuristics take geographic coordinates (lat, lon) and incorporate a longitude correction factor $\lambda = \cos(\phi)$, where ϕ is the average latitude of the graph. This corrects for the fact that one degree of longitude corresponds to fewer meters than one degree of latitude at non-equatorial latitudes (at 45°N, ≈78.5 km vs 111 km).

1) Euclidean distance: The straight-line distance between two points in the coordinate plane, adjusted for longitude distortion:

$$d_E = \sqrt{(\Delta\text{lat})^2 + (\Delta\text{lon} \cdot \lambda)^2} \quad (2)$$

2) Manhattan distance: The sum of the absolute differences along each coordinate axis:

$$d_M = |\Delta\text{lat}| + |\Delta\text{lon}| \cdot \lambda \quad (3)$$

3) Haversine distance: The great-circle distance between two points on a sphere of radius R=6371 km

$$d_H = 2R \cdot \arctan 2\left(\sqrt{a},\ \sqrt{1-a}\right) \tag{4}$$

where $a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right)$ and $R =$ 6371 km. Returns the great-circle distance in km; does not need the λ correction since it handles spherical geometry natively.

D. Scaling

For A* to guarantee optimal paths, the heuristic must be admissible, meaning it must never overestimate the true minimum cost to reach the destination. Since edge weights represent travel time (length divided by speed), the heuristic must be converted into a comparable time estimate.

Euclidean and Manhattan return values in degrees. Because $\lambda = \cos(\phi)$ scales the longitude axis to match the latitude axis, as further discussed in next section, every corrected degree corresponds to approximately $111,320$ m (the length of one degree of latitude). Multiplying by $111,320/v_{max}$ therefore converts the heuristic to seconds, matching the edge weights.

Haversine returns values in kilometers, so it simply needs to be multiplied by $1{,}000/v_{\max}$ to get seconds.

Table I
SCALING FACTORS APPLIED TO EACH HEURISTIC (LONGITUDE CORRECTION $\cos\bar{\phi}$ IS APPLIED INSIDE THE EUCLIDEAN AND MANHATTAN FUNCTIONS).

Heuristic	Scaling factor	Admissible?
Euclidean	1 (no scaling)	Yes (trivially)
Euclidean	$111{,}320/v_{\max}$	Yes
Manhattan	1 (no scaling)	Yes (trivially)
Manhattan	$111{,}320/v_{\max}$	No
Haversine	$1{,}000/v_{\max}$	Yes

Dividing by v_{max}, ensures that the heuristic represents the minimum possible travel time for a certain distance (i.e., traveling in a straight line at the highest speed available); this conservative approach preserves admissibility.

E. Longitude correction

The Euclidean heuristic with longitude correction is admissible: the corrected Euclidean distance underestimates or equals the true great-circle distance, and dividing by $v_{\max}$ (the fastest possible speed) gives a lower bound on travel time.

The Manhattan heuristic, even with the longitude correction, can overestimate: it sums the absolute differences rather than combining them with a square root, so when $\Delta\text{lat} \approx \Delta\text{lon} \cdot \cos\bar{\phi}$, the heuristic can exceed the true straight-line distance by a factor of up to $\sqrt{2}$, it may exceed the true distance, making it inadmissible.

The Haversine heuristic is admissible by construction.

F. Experimental setup

For each city, 10 random origin-destination pairs were generated once and reused across all algorithm configurations. The comparison metric is the number of node expansions (iterations) before reaching the destination.

III. RESULTS

Table II summarizes the average number of iterations across 10 random origin-destination pairs for each algorithm configuration.

Table II
AVERAGE ITERATIONS (NODE EXPANSIONS) OVER 10 RANDOM PAIRS, WITH SPEEDUP RELATIVE TO DIJKSTRA.

Algorithm	Scaling	Turin	Speedup	Aosta	Speedup
Dijkstra	—	6,376	1.00×	347	1.00×
A* Euclidean	1	6,376	1.00×	347	1.00×
A* Euclidean	$111{,}320/v_{\max}$	4,438	1.44×	188	1.85×
A* Manhattan	1	6,376	1.00×	347	1.00×
A* Manhattan	$111{,}320/v_{\max}$	**3,846**	**1.66**×	**154**	**2.26**×
A* Haversine	$1\text{k}/v_{\max}$	4,440	1.44×	188	1.85×

As expected, with no scaling, all heuristics (factor 1) match Dijkstra: the raw degree values ($\sim 10^{-2}$) are negligible compared to edge weights ($\sim 10^0$–10^2), so $f(n) \approx g(n)$ and A* degenerates into Dijkstra..

With proper scaling, the differences become significant.

The **Haversine** heuristic with $1{,}000/v_{\max}$ reduces iterations by 30% on Turin (1.44× speedup) and 46% on Aosta (1.85× speedup) compared to Dijkstra, while maintaining admissibility.

The **Euclidean** heuristic with $111{,}320/v_{\max}$ achieves virtually identical performance (1.44× on Turin, 1.85× on Aosta), which is expected since both compute a comparable approximation of the true straight-line distance when the longitude correction is applied.

The **Manhattan** heuristic with $111{,}320/v_{\max}$ is the most aggressive configuration, achieving a 1.66× speedup on Turin (40% fewer iterations) and a 2.26× speedup on Aosta (56% fewer iterations). However, as discussed in section II-E, it is not admissible: the formula can overestimate the true distance, meaning the returned paths may be suboptimal.

IV. DISCUSSION

A. Unit consistency

Unit consistency between heuristic and edge weights is the determining factor for A* performance, more so than the choice of distance function. Without proper scaling, all three heuristics are effectively useless leading $f(n) \approx g(n)$: A* degeneration into Dijkstra. With it, the differences emerge.

B. Performance and admissibility

Haversine and corrected Euclidean perform almost identically, which is an expected behavior: at the scale of a city ($\sim$10 km), the difference between the Euclidean approximation on a locally flat surface and the true spherical distance is really small. The practical advantage of Haversine is that it requires no manual correction—it handles the degree-to-meter conversion and latitude-dependent scaling internally.

Manhattan achieves the most aggressive pruning but is not admissible with this scaling.

The improvement on Aosta is proportionally larger than on Turin. This is likely due to Aosta's elongated valley geography, which gives the heuristic stronger directional guidance. Turin's as a well-known grid structured cirty offers more alternative routes.

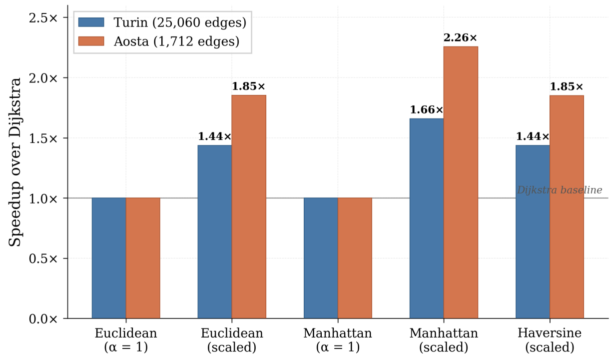

Figure 1. Visual representation of speedup over Dijkstra

Conclusions

This work demonstrates that the choice of heuristic function and its proper scaling are critical for A* performance on real-world road networks. The Haversine and longitude-corrected Euclidean heuristics achieve virtually identical admissible performance ($1.44\times$ speedup on Turin, $1.85\times$ on Aosta), confirming that the flat-Earth approximation is adequate at city-scale distances. The Manhattan distance, while achieving the best speedup ($2.26\times$ on Aosta), remains inadmissible due to the inherent overestimation of the L1 norm. Haversine remains the most suitable heuristic for geographic pathfinding: it is admissible by construction, requires no ad-hoc corrections, provides tight bounds and scale the path finding problem to real-world scenarios that not always rely on city-scale distance. Euclidean distance with longitude correction is a valid alternative at city scale.